

Species associations revealed by an automatic fauna identifier and 1.2 million citizen-science photographs

Draft v1 — 28-sep-2026 · Correlation Project (BioFauna)

Abstract

Marine citizen science has produced photographic observation datasets that are rarely exploited beyond species distributions. We analyse **1,222,170 photographs** and **759,207 geolocated, dated observations** (Minka SDG, iNaturalist and other public sources), automatically identified by **BioFauna** —a k-NN retriever over BioCLIP-2.5 ViT-H/14 embeddings with hierarchical calibration— to uncover **species associations**. We define a sampling *event* (observer × ~1 km cell × day × 3-h window) and evaluate pair co-occurrence with support, spatial and temporal repeatability, finding **70,971 significant pairs**. The method ****recovers known, documented associations* with high robustness** —*e.g. Peltodoris atromaculata on the sponge Petrosia ficiformis** (79 events, lift 11.1, 44 localities, 74 days)— and reveals **undescribed candidates** with plausible biological interpretation, such as jellyfish with commensal hyperiid amphipods, cleaner shrimps with anemones and infaunal species pairs. The work highlights the role of human curators in verifying cryptic species and outlines applications for conservation and outreach.

Keywords: citizen science, interspecific associations, habitat use, deep learning, Mediterranean, underwater photography.

1. Introduction

Amateur underwater photography has become a massive source of biodiversity data. Citizen-science platforms (**Minka SDG**, driven by the **FECIDAS** ecosystem; **iNaturalist**; **GBIF**) accumulate millions of verified observations, and automatic identifiers can consolidate that archive at catalogue scale. Yet most analyses stop at inventories and distribution maps. Photographs also carry **ecological context**: which species occur together, on what substrate and under which conditions. This work explores that latent information.

2. Material and methods

Summary (details in `METODOS.md` and supplementary material):

- **Data:** 1,222,170 photographs; 759,207 observations with coordinates, date and (92 %) time; a curated

catalogue of 2,985 Mediterranean taxa.

- **Identification:** BioFauna (BioCLIP-2.5 ViT-H/14 + FAISS k-NN, k=15, T=0.05 aggregator with a cap of 3

votes per species, hierarchical calibration), with publication guards and leak auditing.

- **Unit of analysis:** $event = (observer \times \sim 1.1 \text{ km cell} \times \text{day} \times 3\text{-h window})$; the set of species seen together.

- **Filters:** catalogue-only; support ≥ 8 events; repeatability across ≥ 3 **localities**, ≥ 3 **days** and ≥ 5 **distinct observers** (the latter added after finding that a single dive trip generates false associations).

- **Metrics:** observed co-occurrence n , expected from marginals and **lift**; Poisson p for screening.

- **Internal validation:** blind search for documented associations (§3.1).

3. Results

3.1 Method validation (documented associations recovered)

association	n	lift	localities	days	source
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<i>Peltodoris atromaculata</i> — <i>Petrosia ficiformis</i>	79	11.1	44	74	classic literature
<i>Felimare picta</i> — <i>Ircinia oros</i>	29	4.1	20	28	host sponge
<i>Cratena peregrina</i> — <i>Eudendrium racemosum</i>	5	10.5	—	—	prey hydroid
~~ <i>Doto paulinae</i> — <i>Sertularella mediterranea</i> ~~	5	48.3	—	—	✗ discarded: single observer (expert-review correction)

3.2 Taxonomic groups

Pairs concentrate in the photographically best-represented families (sparids, blenniids, gobiids, labrids, coralline algae, didemnids, holothurians, actiniids, trochids, dictyotaceans, facelinids, serpulids, sabellids, irciniids). **98 % of pairs are inter-family**, as expected for habitat or trophic associations.

3.3 New candidates (pending expert review)

functional group	A — B	n	lift	localities	days
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|---|---:|---:|---:|---:|---|

commensalism on jellyfish	<i>Blackfordia virginica</i> — <i>Phronima sedentaria</i>	19	653	6	15
cleaning symbiosis	<i>Lysmata grabhami</i> — <i>Telmatactis cricoides</i>	9	547	7	8
infauna (same sediment)	<i>Fustiaria rubescens</i> — <i>Loripinus fragilis</i>	16	892	7	15
infauna	<i>Abra alba</i> — <i>Abra longicallus</i>	9	534	7	9
hard-substrate microhabitat	<i>Macrorhynchia philippina</i> — <i>Telmatactis cricoides</i>	8	765	5	6
gelatinous plankton	<i>Callianira bialata</i> — <i>Vanadis formosa</i>	9	645	7	9

(Full table: supplementary material; main figure: top-12 network.)

3.4 Literature verification of the candidates

After reviewing public literature (WoRMS, Sea Slug Forum, FishBase, JNCC/MarLIN, peer-reviewed journals), pairs are classified into three categories:

Documented (the interaction is already described): *Peltodoris atromaculata*—*Petrosia ficiformis*

(predation), *Felimare picta*—*Ircinia* spp. (diet), *Cratena peregrina*—*Eudendrium racemosum*

(kleptopredation), the floating **Sargassum community** (*Latreutes fucorum*, *Hippolyte coerulescens* and

Scyllaea pelagica), *Phronima sedentaria* as a salp symbiont and *Lampea pancerina* as a salp

predator/parasite.

Plausible (ecologically coherent, no direct citation of the pair): *Lysmata grabhami*—**Telmatactis*

cricoides (*the anemone hosts documented crustacean symbionts* —*Thor amboinensis** in 65 % of anemones and

Stenorhynchus lanceolatus in facultative mutualism— and both species truly co-occur in the Canary

Islands/Madeira), gelatinous-plankton pairs (ctenophores, hyperiids and medusae of the same water mass) and

infaunal sediment pairs (*Abra* spp.; *Fustiaria*—*Loripinus*, both typical of circalittoral soft bottoms).

Dive-trip artefact (species from other oceans photographed together by the same group in one immersion):

Doris fontainii—*Tyrinna delicata* (both from the **South Pacific**, lat −12...−55),

Chromodoris quadricolor—*Hexabranhus sanguineus* (Red Sea/Indo-Pacific), *Abudefduf saxatilis*—

Kyphosus vaigiensis (Atlantic vs. Indo-Pacific; they do not coexist in the Mediterranean). These pairs are

excluded from the ecological analysis of the study area; detecting them is also a useful dataset-quality control signal.

Taxonomic correction: *loripinus_fragilis* (Philippi, 1836, **Lucinidae**) and *limaria_fragilis*

(Gmelin, 1791, **Limidae**) are distinct species; they are not synonyms (WoRMS AphiaID 718970 vs. 216644).

4. Discussion

Candidates fall into three interpretative classes: **commensalism/phoresy** (jellyfish—amphipod), ****cleaning symbiosis (shrimp—anemone) and substrate co-habitation**** (infauna; hard bottoms; coralline algae). The central methodological challenge is separating **interaction** from **coincidence** driven by habitat, season or observer effort; hence the spatial/temporal repeatability requirement and the stratified null currently under development.

The literature review yields two results of general interest. First, **the method recovers already documented associations** (dorid predation, the Sargassum community, gelatinous-zooplankton symbionts), validating the ecological signal against effort noise. Second, classifying the top pairs by the geographic range of their species reveals a **systematic citizen-science artefact**: the highest-lift pairs with extra-Mediterranean species (*Doris fontainii*—*Tyrinna delicata*; *Chromodoris quadricolor*—*Hexabranchus sanguineus*) correspond to **dive trips** by the same group rather than to ecological interactions in the study area. Cross-checking each taxon's geographic range is therefore a necessary quality filter before interpreting any candidate as an ecological association.

5. Limitations

1. Co-occurrence $\neq$ interaction; mitigated by filters and the null model, but directed observation remains the final confirmation.
2. Observer/effort bias (bioblitz-type events).
3. Uneven taxonomic coverage.

6. Conclusions and applications

Citizen-science photographs contain ecological associations retrievable at catalogue scale. Results enable (a) **prioritising** the search for rare species where their associate occurs, (b) **enriching BioQuest** with "if you see X, look for Y" suggestions and per-area association networks, and (c) generating **testable hypotheses** for marine research, with the added value of covering shallow coastal waters where satellites perform worst.

Acknowledgements

To the observer community and, most especially, to the **curators** who sustain the reliability of the identifications—in particular **Miquel Pontes**, **Xavier Salvador** and **Berta Companys**— and to the wider curatorial community of **Minka SDG** and **iNaturalist**. Special mention to **FECDAS** and its **Aneris Project** for **driving marine citizen science and observer training**; to **Minka SDG** and **iNaturalist** for the platforms and the data; and to **GBIF**, **Wikimedia Commons**, **DORIS/FFESSM**, **SeaSlugForum**, **WoRMS** and **FishBase** for images and taxonomic data. This analysis has been possible thanks to the **BioFauna identifier** (BioCLIP-2.5 ViT-H/14 + FAISS) developed in the project.